MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

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Outline

1. Real Analysis Lecture 16
   - Limits of functions
   - Continuity
   - Connectedness

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   - Limits of functions
   - Continuity
   - Connectedness

Limit of a function

Theorem

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Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \to T$ be a function.

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Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \to T$ be a function. Then

$$\lim_{x \to a} f(x) = L$$

Limit of a function

Theorem

Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \to T$ be a function. Then

$$\lim_{x \to a} f(x) = L$$

if and only if

$$\lim_{n \to \infty} f(x_n) = L$$

for all sequences $\{x_n\}$ of values in $A \backslash \{a\}$ which converge to a.

Challenge!

Consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}$$

between metric spaces with the Euclidean metric defined by

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

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Question

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between metric spaces with the Euclidean metric defined by

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Question

What is the limit as $(x, y) \to (0, 0)$?

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

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Then

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Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 0) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} 0 = 0.$$

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Then

$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 1/n) = (0, 0)$$

and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^3}{1/n^4 + 1/n^2} = \lim_{n\to\infty} \frac{1/n}{1/n^2 + 1} = 0.$$

Challenge!

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and

$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^4}{1/n^4 + 1/n^4} = \lim_{n\to\infty} \frac{1}{2} = \frac{1}{2}.$$

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Consider the sequence $\{(x_n, y_n)\}$ defined by

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$$\lim_{n\to\infty}(x_n, y_n) = \lim_{n\to\infty}(1/n, 1/n^2) = (0, 0)$$

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$$\lim_{n\to\infty} f(x_n, y_n) = \lim_{n\to\infty} \frac{1/n^4}{1/n^4 + 1/n^4} = \lim_{n\to\infty} \frac{1}{2} = \frac{1}{2}.$$

The limit DOES NOT EXIST!

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \backslash \{a\}$ which converges to a.

Limit of a function

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($\Rightarrow$) Suppose that $\lim_{x \to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \backslash \{a\}$ which converges to a.
Let $\epsilon > 0$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \backslash \{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.

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Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
Let $\epsilon > 0$.
Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.
Therefore since $x_n \neq a$ for all n, we have that

$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

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$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.

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Proof.

($\Rightarrow$) Suppose that $\lim_{x\to a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A\backslash\{a\}$ which converges to a.
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Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.
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Therefore since $x_n \neq a$ for all n, we have that

$$n \geq N \;\Rightarrow\; 0 < d_S(x_n, a) < \delta \;\Rightarrow\; d_T(f(x_n), L) < \epsilon.$$

Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.
Since $\epsilon > 0$ was arbitrary, this proves $\lim_{n\to\infty} f(x_n) = L$. □

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Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$.

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($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

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To see this, we assume otherwise.

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To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

Limit of a function

Proof.

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To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_)n, L) \geq \epsilon$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A\backslash\{a\}$ which converges to a, we have $\lim_{n\to\infty} f(x_n) = L$. We must show that $\lim_{x\to a} f(x) = L$.

To see this, we assume otherwise.

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In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

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In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

However, $\{f(x_n)\}$ never gets closer than ϵ to L, so it cannot converge to L.

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Proof.

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To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

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In this case, $\{x_n\}$ is a sequence of values in $A\backslash\{a\}$ which converges to a.

However, $\{f(x_n)\}$ never gets closer than ϵ to L, so it cannot converge to L.

This is a contradiction. $\square$

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1. Real Analysis Lecture 16
   - Limits of functions
   - Continuity
   - Connectedness

Continuity

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Definition

Then f is continuous at a point $p \in S$ if if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, p) < \delta \Rightarrow d_T(f(x), f(p)) < \epsilon.$$

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Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$.

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Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$. We say f is continuous on $A \subseteq S$ if $f(x)$ is continuous at every point $p \in A$.

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Then f is continuous at a point $p \in S$ if if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, p) < \delta \Rightarrow d_T(f(x), f(p)) < \epsilon.$$

Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$. We say f is continuous on $A \subseteq S$ if $f(x)$ is continuous at every point $p \in A$. If f is continuous on S, we simply say f is continuous.

Sequence version

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Theorem

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Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function. Then f is continuous at a point p in S if for all sequences $\{x_n\}$ of values in S which converge to p, we have

Sequence version

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function. Then f is continuous at a point p in S if for all sequences $\{x_n\}$ of values in S which converge to p, we have

$$\lim_{n\to\infty} f(x_n) = f(p).$$

Sequence version

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function. Then f is continuous at a point p in S if for all sequences $\{x_n\}$ of values in S which converge to p, we have

$$\lim_{n\to\infty} f(x_n) = f(p).$$

Intuitively, continuity means

$$\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right).$$

Compositions

Theorem

Suppose (S, d_S), (T, d_T), and (U, d_U) are metric spaces and $f : S \to T$ and $g : T \to U$ are functions.

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Theorem

Suppose (S, d_S), (T, d_T), and (U, d_U) are metric spaces and $f : S \to T$ and $g : T \to U$ are functions.
If f is continuous at p and g is continuous at $f(p)$, then the composition $h = g \circ f : S \to U$ defined by $h(x) = g(f(x))$ is continuous at p.

Challenge!

Problem

Prove the previous theorem!

Preimages

Definition

Suppose $f : S \to T$ is a function and that $Y \subseteq T$.

Preimages

Definition

Suppose $f : S \to T$ is a function and that $Y \subseteq T$.
The **inverse image** or **preimage** of Y is the set

$$f^{-1}(Y) = \{x : x \in S, \text{ and } f(x) \in Y\}.$$

Challenge!

Problem

Suppose that $f : S \to T$ and $Y \subseteq Z \subseteq T$.
Prove that

$$f^{-1}(Y) \subseteq f^{-1}(Z).$$

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.
Then f is continuous if and only if $f^{-1}(V)$ is open for all open subsets $V \subseteq T$.

Open sets

Proof.

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$(\Rightarrow)$

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($\Rightarrow$)
Suppose $V \subseteq T$ is open.

Open sets

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($\Rightarrow$)
Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.

Open sets

Proof.

($\Rightarrow$)
Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.
Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.

Open sets

Proof.

($\Rightarrow$)
Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.
Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.
Since f is continuous at x, there exists $\delta > 0$ such that $f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.

Open sets

Proof.

($\Rightarrow$)
Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.
Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.
Since f is continuous at x, there exists $\delta > 0$ such that $f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.
Consequently $B_S(x; \delta) \subseteq f^{-1}(V)$.

Open sets

Proof.

($\Rightarrow$)
Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.
Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.
Since f is continuous at x, there exists $\delta > 0$ such that $f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.
Consequently $B_S(x; \delta) \subseteq f^{-1}(V)$.
This shows that x is an interior point of $f^{-1}(V)$.

Open sets

Proof.

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Suppose $V \subseteq T$ is open.
Then for all $x \in f^{-1}(V)$, we have $x \in S$ and $f(x) \in V$.
Since V is open, there exists $\epsilon > 0$ such that $B_T(f(x); \epsilon) \subseteq V$.
Since f is continuous at x, there exists $\delta > 0$ such that $f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon) \subseteq V$.
Consequently $B_S(x; \delta) \subseteq f^{-1}(V)$.
This shows that x is an interior point of $f^{-1}(V)$.
Since x was arbitrary, this proves $f^{-1}(V)$ is open. $\square$

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($\Leftarrow$)

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($\Leftarrow$)
Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.

Open sets

Proof.

($\Leftarrow$)
Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.

Open sets

Proof.

($\Leftarrow$)
Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.
Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.

Open sets

Proof.

($\Leftarrow$)
Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.
Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.
Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.

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Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.
Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.
Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.
It contains the point x, which must be an interior point.

Open sets

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Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.
Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.
Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.
It contains the point x, which must be an interior point.
This means that there exists $\delta > 0$ such that
$B_S(x; \delta) \subseteq f^{-1}(B_T(f(x); \epsilon))$.

Open sets

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Suppose that for all open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.
Let $x \in S$.
Then for all $\epsilon > 0$, the ball $B_T(f(x); \epsilon)$ is open.
Therefore the preimage $f^{-1}(B_T(f(x); \epsilon))$ is open.
It contains the point x, which must be an interior point.
This means that there exists $\delta > 0$ such that
$B_S(x; \delta) \subseteq f^{-1}(B_T(f(x); \epsilon))$.
Thus $f(B_S(x; \delta)) \subseteq B_T(f(x); \epsilon)$. □

Closed sets

Theorem

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Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \to T$ is a function.
Then f is continuous if and only if $f^{-1}(A)$ is closed for all closed subsets $A \subseteq T$.

Relatively open sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function.
If f is continuous on a subset $A \subseteq S$, then for any open subset $V \subseteq T$ there exists an open subset $U \subseteq S$ with

$$f^{-1}(V) \cap A = U \cap A.$$

Relatively open sets

Proof.

Relatively open sets

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Consider the subspace $(A, d_A = d_S)$ of S.

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The function f restricts to a continuous function $f|_A : A \to T$.

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Consider the subspace $(A, d_A = d_S)$ of S.
The function f restricts to a continuous function $f|_A : A \to T$.
Therefore if $V \subseteq T$ is open, the preimage

$$f|_A^{-1}(V) = \{x : x \in A, \text{ and } f(x) \in V\} = f^{-1}(V) \cap A$$

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Proof.

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The function f restricts to a continuous function $f|_A : A \to T$.
Therefore if $V \subseteq T$ is open, the preimage

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is open in the subspace A of S.

Relatively open sets

Proof.

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The function f restricts to a continuous function $f|_A : A \to T$.
Therefore if $V \subseteq T$ is open, the preimage

$$f|_A^{-1}(V) = \{x : x \in A, \text{ and } f(x) \in V\} = f^{-1}(V) \cap A$$

is open in the subspace A of S.
This is the same thing as

$$f^{-1}(V) \cap A = U \cap A$$

for some open subset

$$U \subseteq S$$

.



Compact sets

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Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ a function.
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Extreme value theorem

Theorem

Suppose (S, d_S) is a metric space and $f : S \to \mathbb{R}$ is a function (to Euclidean space).
If f is continuous on a compact subset $A \subseteq S$, then there exists $a, b \in A$ such that

$$f(a) = \inf f(A), \quad \text{and} \quad f(b) = \sup f(A).$$

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This means that there exists $a, b \in A$ such that $y = f(a)$ and $z = f(b)$. □

Outline

1. Real Analysis Lecture 16
   - Limits of functions
   - Continuity
   - Connectedness

Characterization of intervals

Theorem

If $I \subseteq \mathbb{R}$ is an interval, then there does not exist two nonempty, disjoint open sets $U, V \subseteq \mathbb{R}$ (Euclidean space), with

$$I \subseteq U \cup V.$$

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- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Intermediate value theorem

Theorem (Intermediate Value Theorem)

Suppose $f : [a, b] \to \mathbb{R}$ is a continuous function with the Euclidean metric.
Then for any value y between $f(a)$ and $f(b)$, there exists $c \in [a, b]$ with $f(c) = y$.